Building Machine Learning Models Using Data from BindingDB

Pat Walters | February 24, 2026

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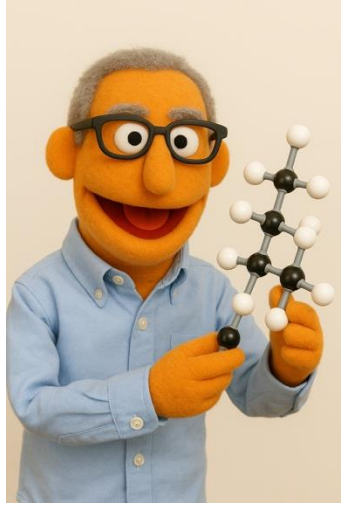
Who Am I?

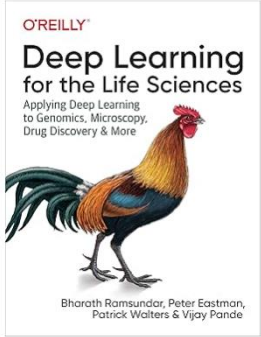


APPLICATIONS OF ARTIFICIAL INTELLIGENCE
IN CONFORMATIONAL ANALYSIS

by
William Patrick Walters

A Dissertation Submitted to the Faculty of the
DEPARTMENT OF CHEMISTRY
in Partial Fulfillment of the Requirements
For the Degree of
DOCTOR OF PHILOSOPHY
in the Graduate College
THE UNIVERSITY OF ARIZONA

1995



Machine Learning

Expert Systems

Lab Automation

Polymer Chemistry

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1980 — 1990 — 2000 — 2010 — 2020

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Here’s What We Will Cover Today

A quick intro to BindingDB

Using BindingDB from Python

- **Converting the BindingDB TSV to a DuckDB database**

Extracting data from BindingDB for a specific protein target

Exploratory data analysis (EDA)

- **Activity distribution**
- **Chemical diversity**
- **Availability of structural information**
- **Assay variability**

Data preprocessing

Data aggregation

ML model building

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BindingDB Contains a Treasure Trove of Chemical and Biological Data

3.2 million experimental affinities (e.g., K_i, IC_{50}, K_d) covering 1.4 million small molecules and 11,000 protein targets

Extensive coverage of U.S. Patents

Rich experimental data

- **Temperature**
- **pH**
- **Buffer composition**

Links to other database such as UniProt, PubMed, ZINC, and PDB

https://bindingdb.org

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BindingDB is Available for Download in Several Formats

BindingDB downloads are available in several formats

• Oracle DB • MySQL DB	Requires a server
• SDF • TSV	Not easily searchable

DuckDB

TSV File → DDB Databases

DuckDB transforms a TSV file into a searchable database

For more information, please visit https://github.com/PatWalters/practical_cheminformatics_tutorials/tree/main/bindingdb

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https://patwalters.github.io

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Pat Walters is Chief Scientist at OpenADMET, an open science initiative that combines insights from high-throughput experimentation, structural biology, and machine learning to improve the prediction of drug absorption, metabolism, excretion, and toxicity. Before his current role, Pat spent thirty years in leadership positions at Relay Therapeutics and Vertex Pharmaceuticals. Pat is the 2023 recipient of the Herman Skolnik Award for Chemical Information Science from the American Chemical Society. He is a member of the editorial advisory boards for the Journal of Chemical Information and Modeling and Artificial Intelligence in the Life Sciences, and previously held a similar role with the Journal of Medicinal Chemistry. Pat is co-author of the book "Deep Learning for the Life Sciences", published in 2019 by O'Reilly and Associates. He received his Ph.D. in Organic Chemistry from the University of Arizona where he studied the application of artificial intelligence in conformational analysis. Prior to obtaining his Ph.D., Pat worked at Varian Instruments as both a chemist and a software developer. He received his B.S. in Chemistry from the University of California, Santa Barbara.

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